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EXP: 1.1

```
clc; clear; close all;
```

```
Fs = 10000;
```

```
Fp = 1000;
```

```
Fst = 1500;
```

```
Rp = 1;
```

```
Rs = 60;
```

```
Wp = Fp/(Fs/2);
```

```
Ws = Fst/(Fs/2);
```

```
[N,Wn] = buttord(Wp,Ws,Rp,Rs);
```

```
fprintf('Minimum Filter Order (N) = %d\n',N);
```

```
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```

```
[b,a] = butter(N,Wn,'low');
```

```
figure;
```

```
freqz(b,a,1024,Fs);
```

```
title('Magnitude and Phase Response - LPF');
```

```
figure;
```

```
[h,n] = impz(b,a,50);
```

```
stem(n,h,'filled'); grid on;
```

```
title('Impulse Response'); xlabel('n'); ylabel('Amplitude');
```

```
figure;
```

```
stepz(b,a,50);
```

```
title('Step Response');
```

```
figure;
```

```

grpdelay(b,a,1024,Fs);

title('Group Delay');

figure;

zplane(b,a);

title('Pole-Zero Plot - LPF');

t = 0:1/Fs:0.05;

x = sin(2*pi*500*t) + sin(2*pi*3000*t); % 500 Hz (pass) + 3000 Hz (reject)

y = filter(b,a,x);

figure;

subplot(2,1,1); plot(t,x); grid on;

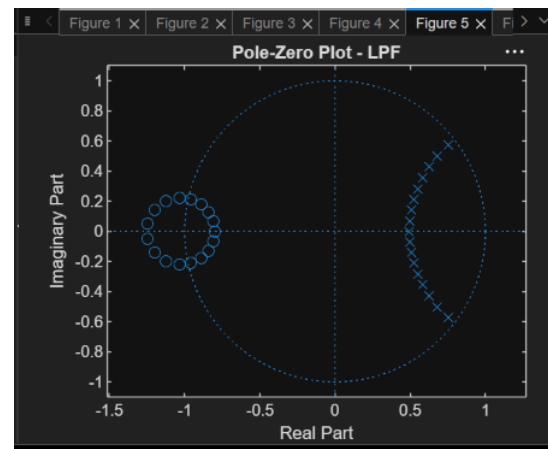
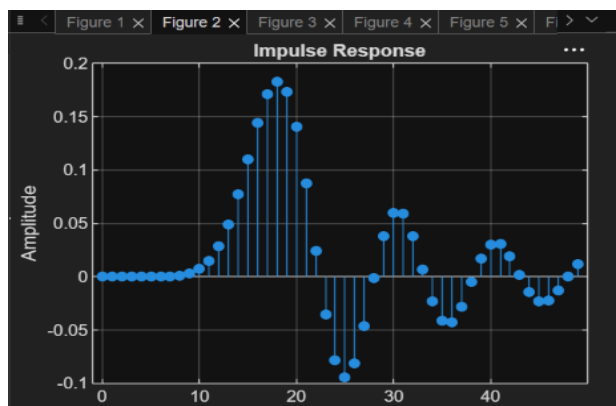
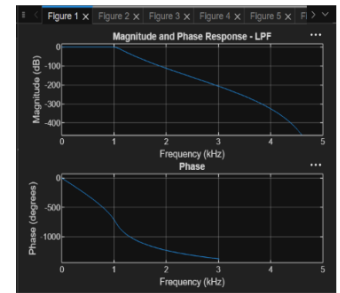
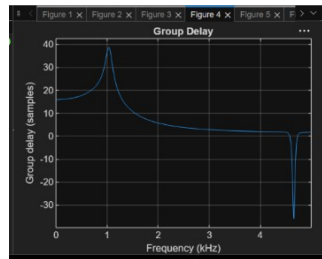
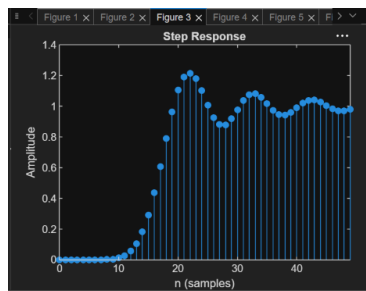
title('Input Signal'); xlabel('Time (s)'); ylabel('Amplitude');

subplot(2,1,2); plot(t,y); grid on;

title('Filtered Output Signal (Low-Pass)'); xlabel('Time (s)'); ylabel('Amplitude');

```

OUTPUT:



EXP: 1.2:

CODE:

```
clc; clear

Fs=10000; Fp=3000; Fst=2000;

[N,Wn]=buttord(Fp/(Fs/2),Fst/(Fs/2),1,60); [b,a]=butter(N,Wn,'high');

freqz(b,a)

% Impulse Response

figure

[h,n]=impz(b,a,50);

stem(n,h), grid on

title('Impulse Response')

figure

stepz(b,a)

title('Step Response')

figure

grpdelay(b,a)

title('Group Delay')

figure

zplane(b,a)

title('Pole-Zero Plot')

fprintf('Minimum Filter Order (N) = %d\n',N); fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```

OUTPUT:

